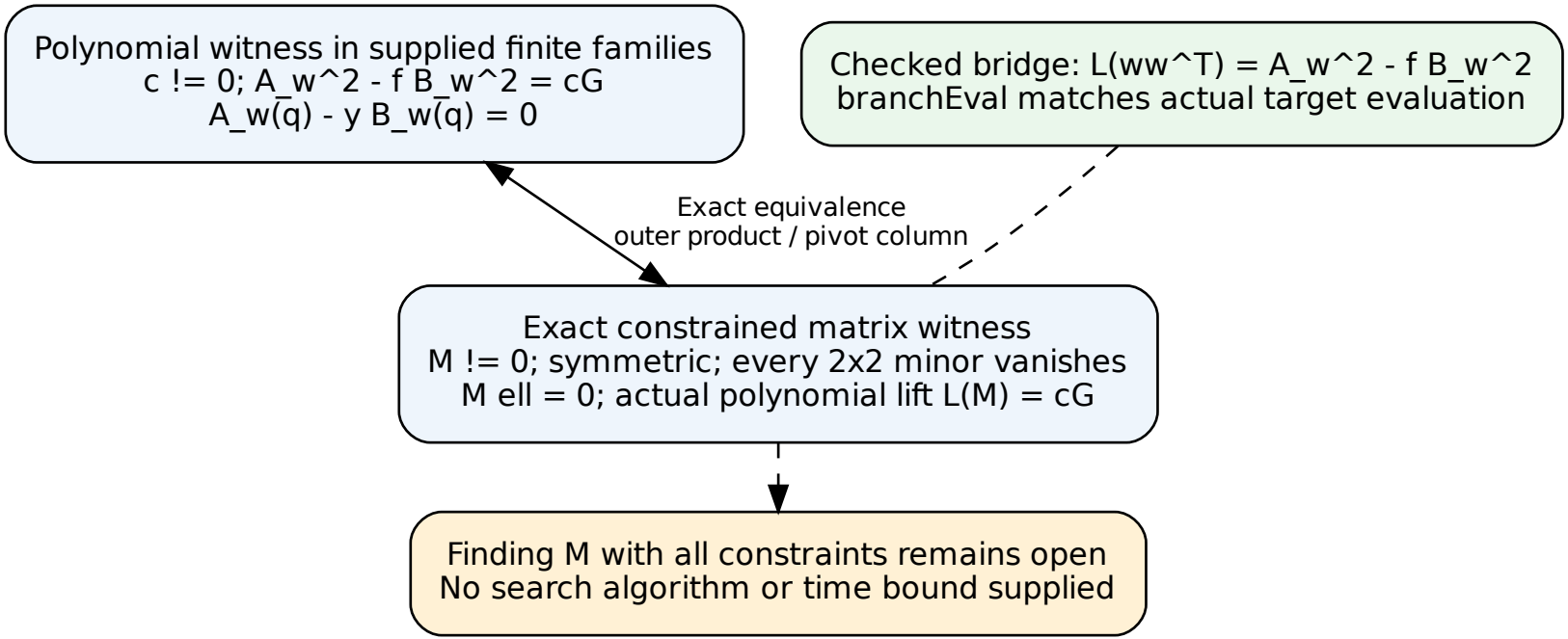
Polynomial witness in supplied finite families
c != 0; A_w^2 - f B_w^2 = cG
A_w(q) - y B_w(q) = 0
Checked bridge: L(ww^T) = A_w^2 - f B_w^2
branchEval matches actual target evaluation
Exact equivalence
outer product / pivot column
Exact constrained matrix witness
M != 0; symmetric; every 2x2 minor vanishes
M ell = 0; actual polynomial lift L(M) = cG
Finding M with all constraints remains open
No search algorithm or time bound supplied